CS6660: Mathematical Foundations of Data Science (Probability)

Practice Problems 04

భారతీయ సాంకేతిక విజ్ఞాన సంస్థ హైదరాబాద్
भारतीय प्रौद्योगिकी संस्थान हैदराबाद
Indian Institute of Technology Hyderabad

1. For any $x \in \mathbb{R}$, let $\lfloor x \rfloor$ denote the largest integer lesser than or equal to x.
   Thus, for instance, $\lfloor 3.5 \rfloor = 3$, $\lfloor -8.9 \rfloor = -9$, $\lfloor 2 \rfloor = 2$, and so on.
   Suppose that $X \sim$ Exponential(1). From first principles, determine the distribution of $Y = \lfloor X \rfloor$.

2. Let X be a discrete random variable with the following PMF:

$$p_X(x) = \begin{cases} 0.1, & x = 10, \\ 0.2, & x = 20, \\ 0.3, & x = 30, \\ 0.4, & x = 40, \\ 0, & \text{otherwise.} \end{cases}$$

   (a) Compute the CDF F_X and the inverse CDF function F_X^{-1} (refer to the lecture notes for the exact formula of the inverse CDF function) of the random variable X.

   (b) Find the distribution of $Y = F_X(X)$.

   Hint: Identify the values taken by the random variable Y and the corresponding probabilities.

3. Let $X_1, X_2 \overset{\text{i.i.d.}}{\sim}$ Exponential(λ).

   (a) Using the bivariate Jacobian transformation formula, compute the joint PDF of $Y_1 = X_1 + X_2$ and $Y_2 = X_1/X_2$, and determine if Y_1 is independent of Y_2..

   (b) Determine the conditional PDF of Y_1, conditioned on the event $\{Y_2 = 1\}$.

   (c) Determine the joint PDF of Y_1 (as defined above) and $Y_3 = X_1 - X_2$.

   (d) Determine the conditional PDF of Y_1, conditioned on the event $\{Y_3 = 0\}$.

   (e) Observe that the event $\{Y_3 = 0\}$ is same as the event $\{Y_2 = 1\}$. Yet, are the conditional PDFs in parts (b) and (d) identical? If not, what could be a possible reason for this?

4. Let X, Y be jointly continuous with the joint PDF

$$f_{X,Y}(x,y) = \begin{cases} \dfrac{1}{x}, & 0 \le y \le x \le 1, \\ 0, & \text{otherwise.} \end{cases}$$

   (a) Determine the PDF of $Z = X + Y$.

   (b) Compute $\mathbb{P}(\{Z \le 1\})$.

5. Numbers from $[0, 1]$ are picked uniformly, independently, and sequentially over time.
   Let X_n denote the number picked at time n, where $n \in \{0, 1, 2, \ldots\}$.
   Let N be the random variable defined as

$$N = \min\{n \ge 1 : X_n < X_0\}.$$

   That is, N denotes the first time index $n \ge 1$ at which the value of X_n goes below the value of X_0.

   (a) For any fixed $n \in \mathbb{N}$, determine $\mathbb{P}(\{N = n\})$.

   Hint: The event that $N = n$ is identical to the event that

$$X_1 \ge X_0 \quad \text{and} \quad X_2 \ge X_0 \quad \text{and} \quad \cdots \quad \text{and} \quad X_{n-1} \ge X_0 \quad \text{and} \quad X_n < X_0.$$

   (b) Compute $\mathbb{P}(\{N > 2\})$.

6. **(Memoryless Property)**

(a) Show that if $X \sim$ Geometric(p) for some $p < 1$, then it satisfies the following memoryless property:

$$\mathbb{P}(\{X > k + n\} \mid \{X > k\}) = \mathbb{P}(\{X > n\}) \qquad \forall k, n \in \mathbb{N}.$$

(b) Show that if $X \sim$ Exponential(μ), then it satisfies the following memoryless property:

$$\mathbb{P}(\{X > t + s\} \mid \{X > t\}) = \mathbb{P}(\{X > s\}) \qquad \forall s, t > 0.$$